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Ontology-Driven Decision Support for Machine Learning Method Selection in Product Development and Production

Combining Semantic Literature Retrieval with Ontology-Based Filtering and Implementation Support

Master's thesis in Engineering and ICT
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ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **NTNU** Norwegian University of Science and Technology

CHAPTER

ONE

INTRODUCTION

1.1 Motivation

- Increasing use of machine learning in product development and production
- Difficulty in selecting suitable ML methods for specific problems
- Gap between research literature and practical implementation
- Need for structured decision support
- Potential value of combining ontology and literature-based retrieval
- Relevance for engineers and practitioners with limited ML expertise

1.2 Pre-master Project Work

- Prior work:
  - semi-structured analysis of a large set of articles
  - development of an article-based roadmap
- Transition:
  - need to formalize and structure knowledge
- Current thesis:
  - extends previous work
  - introduces ontology, embeddings, and system implementation
  - aims to automate and generalize the workflow

1.3 Problem Description

- Users struggle to identify relevant ML methods for their problem
- Existing approaches are either too generic or too technical
- Lack of tools linking:
  - problem descriptions
  - ML methods
  - supporting research articles
- Challenge of utilizing large volumes of literature effectively
- Need for:
  - interpretable recommendations
  - traceability to literature
  - support for early implementation steps
- Context: system combining ontology + embeddings + article retrieval

1.4 Research Questions

- How can ontology-based knowledge improve ML method selection?
- How can semantic similarity between problem descriptions and literature support recommendations?
- How can a tool combine structured knowledge and unstructured text effectively?
- To what extent can such a system support early-stage ML implementation?

1.5 Thesis Structure

- Chapter 1: Introduction
- Chapter 2: Background and state of the art
- Chapter 3: Methodology
- Chapter 4: Implementation
- Chapter 5: Results
- Chapter 6: Discussion
- Chapter 7: Conclusion and future work

CHAPTER

TWO

BACKGROUND AND STATE OF THE ART

2.1 Machine Learning Method Selection

- Overview of ML paradigms (supervised, unsupervised, etc.)
- Typical ML tasks (classification, regression, time series, etc.)
- Challenges in selecting appropriate methods
- Existing guidelines or frameworks for method selection

2.2 Decision Support Systems

- Systems that support decision-making, not replace it
- Knowledge-based vs data-driven systems
- Often interactive and require user input
- Relevance for engineering and industrial contexts

2.3 Implementation Support for Machine Learning

- Selecting a method is only the first step in ML projects
- Users often need support to start implementation
- Templates and reusable workflows can reduce effort
- Such support can lower the barrier for non-experts

2.4 Ontologies and Knowledge Representation

- What an ontology is (concepts, relations, structure)
- Use of ontologies in engineering and ML domains
- Benefits: structure, explainability, reasoning
- Limitations: manual effort, rigidity

2.5 Semantic Retrieval and Text Similarity

- Text can be represented as embeddings (vectors)
- Similarity measures can compare meaning between texts
- Used to match problem descriptions with relevant content
- More flexible than keyword-based search

2.6 Literature-Based Discovery / Retrieval

- Scientific articles are a rich but unstructured knowledge source
- Challenges include large volume, noise, and unclear relevance
- Methods can be linked to articles through extraction and metadata
- Semantic retrieval can improve matching between problems and literature

2.7 Human-in-the-Loop and Feedback

- User input helps guide and refine recommendations
- Feedback can improve system performance over time
- Requires large user base and careful design to avoid bias

2.8 Web-Based ML Tools / Interfaces

- User interfaces allow interaction with the system
- Should support clear input and understandable output
- Transparency helps users trust recommendations
- Focus is on usability, not detailed UX design

2.9 ML in Product Development and Production

- Go through findings from the literature review in pre-master project
- ML is used for tasks such as Quality prediction and defect detection, Process optimization, Predictive maintenance, Time series forecasting
- Product development often has limited and uncertain data
- Production systems may have structured and continuous data
- Selecting suitable ML methods is challenging in these contexts

CHAPTER THREE

METHODOLOGY

CHAPTER

FOUR

IMPLEMENTATION

CHAPTER FIVE

RESULTS

CHAPTER

SIX

DISCUSSION

CHAPTER

SEVEN

CONCLUSIONS

APPENDICES